Cosmological Memory of a Gaussian Detector in an Expanding Universe: Finite-N Dynamics and the Breakdown of the Markovian Approximation in FRW

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Abstract

We present a 3+1D Wightman-based memory analysis of a Gaussian detector coupled to a scalar field through a matter-dominated expansion history. The calculation is framed via the modular flow of Tomita–Takesaki and the quantum Fisher information F_Q of the detector's reduced state, testing the intuition of the Universal Applied Time (UAT) framework that time is a measure of causal connection.

Three results are reported. First, the Unruh–DeWitt consistency check (KMS condition) passes at machine precision ($\max|\Delta W/W| \lesssim 10^{-14}$ for accelerations $a \in \{0.5, 1.0, 2.0\}$), which validates the numerical apparatus. Second, the maximum information mutual $I(O{:}\phi)_{\max}$ between detector and field converges with N to approximately 0.26 nats, while the power-law exponent α of the Fisher-information decay does *not* converge with N. We show that this non-convergence is a physical consequence of the finite-dimensional unitary dynamics (quasi-periodic orbits, no genuine decay), not a numerical artefact.

Third, we compare two continuum models. A *static-bath* quantum Brownian motion model with Ohmic spectral density gives $\Gamma = \pi\eta/2$, independent of temperature and cutoff. However, an *exact* treatment using the Wightman function of a scalar field in FRW shows that the effective damping rate γ_{eff} decays with the expansion and exhibits memory effects (sign-changing oscillations, associated with particle creation) in the matter and dark-energy regimes. The global decay is robust against the choice of UV regulator; the precise location of the sign change is scheme-dependent. The Ohmic closure is mathematically exact but describes the wrong system: a detector coupled to an external static bath, not to a cosmological field.

1 Introduction

The Universal Applied Time (UAT) framework—also referred to by its Spanish acronym TAU, *Tiempo Aplicable Unificado*—was developed independently by the author during 2025 and proposed that the time assigned to an event is a measure of the causal connection between that event and an observer. The original formulation expressed this intuition through additive equations with multiplicative factors encoding cosmological, relativistic, and quantum-gravitational corrections. Systematic tests against DESI DR2 baryon acoustic oscillation data and Pantheon supernova data established that the additive formalism was not competitive with ΛCDM, and subsequent analysis identified the root cause as a structural incompatibility between the relationalist intuition and the additive formalism (4).

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The present paper does not attempt to salvage the original equations. Instead, it takes the underlying intuition seriously and asks what mathematical object it actually points to. The answer—developed in collaboration with automated reasoning tools and formalised in a companion specification—is that the intuition reduces, in the calculable regime, to a statement about the *informational cost* of maintaining temporal resolution in an expanding universe.

It is well established that the Markovian approximation for quantum detectors fails in curved spacetimes (19; 20; 21; 23), and that non-Markovianity and memory effects are standard features of open quantum systems (24; 25). In particular, **(author?)** (22) have shown that an Unruh–DeWitt detector with harmonic oscillator internal degrees of freedom, coupled to a quantum field in an expanding universe, exhibits non-Markovian memory effects associated with the squeezing of the field by the cosmic dynamics. The contribution of the present work is not to claim the first demonstration of non-Markovianity in cosmology. It is to provide a specific 3+1D Gaussian-detector calculation across a full matter-dominated expansion history, framed via the quantum Fisher information and the modular flow, and to compare explicitly against the Ohmic quantum Brownian motion closure, showing that the latter is exact for the wrong system.

1.1 Relation to the modular program

The technical framework of this paper is the modular theory of Tomita–Takesaki (5; 6), as applied to quantum clocks in cosmological settings by recent work (8; 9; 10; 11; 12; 13; 14).

The UAT/TAU framework was developed independently by the author in 2025, from a starting point in cosmological numerical modeling and direct communication with experimental collaborations. Its public development record is documented in Section 11. For the reader's reference, the following dates are relevant:

- Chen and Penington formalised the modular treatment of cosmological clocks in June 2024 (9).
- De Vuyst and collaborators developed the crossed-product and quantum-reference-frame formulation of gravitational entropy in December 2024 (12).
- Speranza proposed the intrinsic cosmological observer in April 2025 (11).
- The UAT/TAU framework was released publicly on GitHub in June 2025 and on Zenodo in November 2025 (Section 11).
- Chen and collaborators extended the crossed-product approach to covariant observers in de Sitter space in November 2025 (10).

The convergence of UAT with the modular program is therefore independent and parallel. The framework does not claim priority over **(author?)** (9, 12), whose work predates it. What UAT contributes is a phenomenological intuition developed outside the algebraic quantum field theory community, from a starting point in cosmological numerical modeling. The present paper takes that intuition, translates it into the language of modular theory, and provides its first quantitative test.

2 What UAT/TAU Really Points To

Before the technical development, we state the interpretive claim in one sentence, because it is the only claim this paper endorses:

> **The capacity of a quantum observer to resolve cosmic time is governed by its mutual information with the field.**

This is the honest translation of the original UAT/TAU intuition to the language of modular theory and quantum information. It replaces the earlier language of "applied time" and "causal connection" by a quantity that is mathematically well defined, computable, and falsifiable in principle: the quantum Fisher information of the detector's reduced state with respect to the cosmic time parameter.

No stronger claim is made in this paper. In particular, we do not assert that time "runs differently" for different observers, nor that the metric is modified, nor that any cosmological parameter is altered. The observable is the resolution of a clock, not the rate of time.

3 Model

3.1 Background and field

We work in a spatially flat FRW universe with line element

$$ds^2 = -dt^2 + a^2(t)\, d\vec{x}^2, \tag{1}$$

and adopt the matter-dominated regime $a(\eta) = \eta^2$ in conformal time η, which is sufficient for the numerical demonstration. A real massive scalar field ϕ of mass $m_\phi \ll H_0$ is expanded in comoving modes,

$$\phi(\vec{x}, \eta) = \frac{1}{\sqrt{L^3}} \sum_{\vec{k}} \left[a_{\vec{k}} \phi_k(\eta) e^{i\vec{k}\cdot\vec{x}} + \text{h.c.} \right], \tag{2}$$

with the mode functions satisfying the standard FRW equation. The field is prepared in the Bunch–Davies vacuum at early times.

3.2 Detector

The observer is modelled as a harmonic oscillator of frequency ω_0 with canonical quadratures $R_O = (x_O, p_O)$ obeying $[x_O, p_O] = i$ ($\hbar = 1$). We use a harmonic oscillator rather than a two-level system because the bilinear coupling

$$H_{\text{int}} = g\, x_O\, \phi(\vec{x}_O, \eta) \tag{3}$$

preserves Gaussianity of the joint state, reducing the entire problem to matrix algebra on covariance matrices. A two-level detector coupled through σ_x would break Gaussianity and would in addition be subject to the bound $I(O{:}\phi) \leq \ln 2$, which truncates the analysis prematurely.

3.3 Covariance matrix formalism

The joint state is Gaussian and fully characterised by the covariance matrix

$$\sigma_{ij} = \tfrac{1}{2} \langle \{ R_i - \langle R_i \rangle, R_j - \langle R_j \rangle \} \rangle, \tag{4}$$

with $R = (x_O, p_O, x_1, p_1, \ldots, x_N, p_N)$ for N field modes retained. The symplectic eigenvalue $\nu_m \geq 1/2$ of each 2×2 block determines the von Neumann entropy

$$S(\nu) = (\nu + \tfrac{1}{2}) \ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2}) \ln(\nu - \tfrac{1}{2}). \tag{5}$$

The information mutual between the detector and the field is

$$I(O{:}\phi) = S(\sigma_O) + S(\sigma_\phi) - S(\sigma_{\text{total}}). \tag{6}$$

3.4 Quantum Fisher information

The Fisher information of the detector state with respect to a time parameter λ is, for a Gaussian state,

$$F_Q^{(\lambda)} = \tfrac{1}{2}(\partial_\lambda \sigma)^T (\sigma \otimes \sigma - \tfrac{1}{4}\Omega \otimes \Omega)^{-1} (\partial_\lambda \sigma), \tag{7}$$

where Ω is the symplectic form and the derivative is taken with respect to λ. The Cramér–Rao bound then gives the resolution limit

$$\Delta\lambda \geq \frac{1}{\sqrt{F_Q^{(\lambda)}}}. \tag{8}$$

Two distinct time parameters are physically relevant:

- the detector's proper time τ_O, and
- the cosmic (conformal or scale-factor) time η or t.

For an expanding universe, $dt = a\, d\eta$, so $F_Q^{(t)} = F_Q^{(\eta)}/a^2$. Part of any apparent degradation of clock resolution is therefore purely kinematic. Any genuine dynamical effect must be identified after removing this factor.

4 Unruh–DeWitt Consistency Check

Before computing in FRW, we verify the apparatus against the canonical result for an accelerated detector in Minkowski space. The Wightman function along a uniformly accelerated trajectory in $1+1$ dimensions is

$$W(\Delta\tau) = -\frac{1}{4\pi} \ln\left[-\frac{4}{a^2} \sinh^2\left(\frac{a\,\Delta\tau}{2} \right) - i\varepsilon \right]. \tag{9}$$

The KMS condition for a thermal state at inverse temperature $\beta = 2\pi/a$ requires

$$W(\Delta\tau - i\beta) = W(-\Delta\tau). \tag{10}$$

We test this identity numerically for $a \in \{0.5, 1.0, 2.0\}$ and $\tau \in [0.01, 5.0]$. The results are reported in Table 1.

a	$T = a/2\pi$	$\max \lvert \Delta W/W \rvert$
0.5	0.0796	1.0×10^{-14}
1.0	0.1592	5.0×10^{-15}
2.0	0.3183	2.6×10^{-15}

Table 1: KMS condition test. The maximum relative deviation is at the level of double-precision round-off. The apparatus is correct.

The KMS test passes at machine precision. This confirms that the detector-field formalism, the Wightman function, and the numerical implementation are correct.

5 Results in a Finite Truncation

5.1 Setup

We take N field modes with comoving momenta $k_j = 2\pi j/L$, $j = 1, \ldots, N$, on a comoving box of size $L = 2\pi$. The detector frequency is $\omega_0 = 1$, the coupling is $g = 1$, and the field mass is $m_\phi = 0.25$. The conformal time runs from $\eta_i = 2$ to $\eta_f = 7$, corresponding to $a \in [4, 49]$ in the matter-dominated background $a(\eta) = \eta^2$. The joint system is evolved unitarily with the symplectic propagator, from the vacuum state $\sigma(0) = I/2$.

5.2 Information mutual and its convergence

The maximum information mutual between detector and field during the evolution is reported in Table 2.

N	$\nu_{\max}$	$I(O{:}\phi)_{\max}$	α (envelope fit)
5	0.5252	0.236	-0.04
10	0.5269	0.249	$+0.42$
20	0.5277	0.255	-1.02
40	0.5283	0.259	-1.49
80	0.5285	0.261	-0.66

Table 2: Convergence of observables with the number N of retained field modes. The information mutual $I(O{:}\phi)_{\max}$ converges smoothly. The power-law exponent α of the Fisher information does not converge and changes sign.

Two facts emerge:

- $I(O{:}\phi)_{\max}$ converges monotonically to approximately 0.26 nats as N increases. This is a stable, well-defined observable of the model.
- The exponent α fitted to the envelope of $F_Q^{(\eta)}$ in the window $a \in [15, 35]$ does not converge. It changes sign between successive values of N.

5.3 Interpretation of the non-convergence of α

The non-convergence of α is not a numerical artefact. It is a physical consequence of the finite-dimensional, unitary dynamics of the model: the joint system has $2(N+1)$ degrees of freedom, its spectrum is discrete, and unitary evolution produces quasi-periodic orbits, not monotonic decay. The apparent "decay" of $F_Q^{(\eta)}$ in any finite window is an interference pattern, not a genuine power-law. Any genuine decay requires a continuum limit.

6 A Static-Bath Model: Quantum Brownian Motion

The simplest continuum model is to replace the discrete set of N modes by a fixed Ohmic spectral density. This is *not* derived from the Wightman function of a field in FRW; it is a phenomenological closure, equivalent to coupling the detector to an external bath at fixed temperature T_{eff}.

6.1 Spectral density and master equation

We adopt an Ohmic spectral density with exponential ultraviolet cutoff,

$$J(\omega) = \eta\, \omega\, e^{-\omega/\Lambda}, \tag{11}$$

where η is a dimensionless coupling constant and Λ is the cutoff frequency. In the Markovian regime, the reduced dynamics of the detector is described by the quantum Brownian motion master equation in the high-temperature limit,

$$\frac{d\sigma}{d\eta} = M\sigma + \sigma M^T + N, \tag{12}$$

with drift matrix

$$M = \begin{pmatrix} 0 & 1/m \\ -m\omega_0^2 & -\gamma/m \end{pmatrix}, \tag{13}$$

and quantum noise matrix

$$N = \begin{pmatrix} 0 & 0 \\ 0 & m\gamma\omega_0 \coth\left(\frac{\omega_0}{2T_{\rm eff}}\right) \end{pmatrix}. \tag{14}$$

The damping coefficient is $\gamma = \pi\eta/2$. The model relaxes to $\nu_\infty = \frac{1}{2}\coth(\omega_0/2T_{\rm eff})$, and the symplectic eigenvalue decays with rate $\Gamma = \gamma = \pi\eta/2$. This is mathematically exact for the Ohmic bath.

7 Exact FRW Treatment: Bath Dilution and Memory

The correct way to obtain the damping kernel for a Gaussian detector coupled to a scalar field in FRW is to compute it from the Wightman function of the field.

7.1 Wightman function and damping kernel

For a massless scalar field in a spatially flat FRW universe with $a(\eta) = \eta^n$ and in the Bunch–Davies vacuum, the rescaled mode functions are

$$u_k(\eta) = \sqrt{\frac{\pi|\eta|}{4}}\, H_\nu^{(2)}(k|\eta|), \qquad \nu = \left|n - \tfrac{1}{2}\right|, \tag{15}$$

and the physical field is $\phi_k = u_k/a(\eta)$. We define the Wightman function as

$$W(\eta,\eta') = \frac{1}{2\pi^2}\,\frac{1}{a(\eta)a(\eta')}\int_0^\infty dk\, k^2\, u_k(\eta)\, u_k^*(\eta'). \tag{16}$$

The damping kernel that enters the non-Markovian equation of motion for the detector is

$$\gamma(\eta,\eta') = 2g^2\, \mathrm{Im}\, W(\eta,\eta')\, \theta(\eta-\eta'). \tag{17}$$

The effective damping rate at time η is

$$\gamma_{\rm eff}(\eta) = \int_0^{\eta-\eta_{\rm min}} d\tau\, \gamma(\eta,\eta-\tau)\cos(\omega_0\tau), \tag{18}$$

with $\eta_{\rm min} \sim 1/\omega_0$.

7.2 Analytic cases and sign convention

In two cases, closed-form expressions are available. For radiation ($n = 1$), $u_k'' + k^2 u_k = 0$ (since $a''/a = 0$), and with the Bunch–Davies convention $u_k \sim e^{-ik\eta}$ for $k|\eta| \gg 1$,

$$\mathrm{Im}\, W_{\rm conf}(\eta,\eta') = -\frac{1}{4\pi^2 a(\eta)a(\eta')}\,\frac{2\Delta\varepsilon}{(\Delta^2+\varepsilon^2)^2}, \qquad \Delta = \eta-\eta'. \tag{19}$$

The negative sign follows from $\mathrm{Im}[e^{-ik\eta}e^{+ik\eta'}] = -\sin(k\Delta)$ for $\Delta > 0$. The corresponding $\gamma_{\rm eff}$ is negative, which in the convention of (18) corresponds to positive damping. This is exact and has no infrared divergence.

For de Sitter minimal ($n = -1$), we use the expanding branch $\eta < 0$ with scale factor $a(\eta) = -1/(H\eta)$. The Bunch-Davies vacuum corresponds to the positive-frequency solution $\sim e^{-ik\eta}$ in the far past. With this convention, the analytic expression is

$$\mathrm{Im}\, W_{\rm dS}^{\rm bare} = -\frac{1}{4\pi^2}\left[\frac{2\eta\eta'\varepsilon\Delta}{(\Delta^2+\varepsilon^2)^2} + \arctan\left(\frac{\Delta}{\varepsilon}\right) - \frac{\Delta\varepsilon}{\Delta^2+\varepsilon^2}\right]. \tag{20}$$

The bare value is negative for $\Delta > 0$ (anti-damping). The sign is fixed by the Bunch-Davies choice of $H^{(2)}$; it is not an artefact of the regulator. The arctan term dominates at late times and is the infrared remnant.

For matter ($n = 2$), the k-integral is computed numerically with a UV regulator. In all cases, we have verified that the qualitative behaviour (decay, sign change) is robust against the choice of regulator, as discussed next.

7.3 Robustness against UV regulators

The specific numerical values of γ_{eff} depend on the UV regulator ε and on the numerical integration parameters. To verify that the qualitative conclusions do not depend on this choice, we have computed γ_{eff} using two smooth regulators: (i) exponential $e^{-\varepsilon k}$ and (ii) Gaussian $e^{-\varepsilon^2 k^2}$.

A third regulator (sharp step at $k_{\max}$) was tested during development but excluded because it introduces Gibbs oscillations in the oscillatory k-integral, rendering the result numerically unstable. The oscillatory factor $u_k u_k^*$ combined with a discontinuous cutoff produces spurious high-frequency modes that dominate the τ-integration. This is a known numerical artefact, not a physical effect.

The results for the two smooth regulators are reported in Table 3. Two features are robust:

- **Robust: global decay.** In all backgrounds and for both regulators, $|\gamma_{\text{eff}}|$ decreases with η and approaches zero at late times. The static-bath prediction $\Gamma = \pi g^2/2$ is not reproduced.
- **Robust: failure of Markovian dissipation.** In matter, γ_{eff} changes sign for both regulators, indicating memory effects (information returns to the detector from the field via particle creation). A Markovian dissipation term, which is by construction positive, cannot reproduce this.

Background	Regulator	γ_{eff} (early)	γ_{eff} (late)
Radiation	exp	Negative	Negative (decayed)
Radiation	Gaussian	Negative	Negative (decayed)
Matter	exp	Positive	Sign change
Matter	Gaussian	Positive	Sign change
de Sitter	exp	Negative (large)	Negative (decayed)
de Sitter	Gaussian	Negative (large)	Negative (decayed)

Table 3: Robustness of the effective damping rate against the choice of smooth UV regulator. The global decay and the failure of Markovian dissipation are invariant across regulators and across backgrounds. The magnitudes depend on the regulator and are not quoted as physical predictions.

7.4 Results

We have computed $\gamma_{\text{eff}}(\eta)$ for the three backgrounds. The specific numerical values of γ_{eff} depend on the UV regulator, on the infrared subtraction scheme, and on the numerical integration parameters. They are therefore *not* quoted here as physical predictions. What is robust across all choices is the qualitative behaviour, summarized in Table 4.

Three features are robust:

1. $|\gamma_{\text{eff}}|$ decreases with η rather than approaching a constant. The static-bath prediction $\Gamma = \pi g^2/2$ is not reproduced.

Background	n	γ_{eff} behaviour	Notes
Radiation	1	Negative, decays with η	Exact conformal case; positive damping
Matter	2	Oscillatory, sign change	Memory from particle creation; non-Markovian
de Sitter	-1	Negative (anti-damping)	IR sensitive to subtraction; no constant limit

Table 4: Qualitative behaviour of the effective damping rate from the exact FRW Wightman function. In no case does γ_{eff} approach the static-bath prediction $\pi\eta/2$. The qualitative behaviour is robust across smooth regulator choices (see Table 3). The sign convention follows Bunch–Davies positive frequency, see Section 7.2.

2. In the matter regime, γ_{eff} exhibits sign-changing oscillations, indicating memory effects (information returns to the detector from the field via particle creation). A Markovian dissipation term, which is by construction positive, cannot reproduce this.

3. In de Sitter, the massless minimally coupled field has an infrared divergence, but in all smooth regularization schemes γ_{eff} decays as $\eta \to 0^-$ rather than approaching a constant.

8 Comparison and Cosmological Implications

The two continuum models give qualitatively different answers: the static bath gives exponential relaxation with $\Gamma = \pi\eta/2$ independent of $a(\eta)$, while the FRW field gives suppression with the expansion plus oscillatory memory effects. Only the second corresponds to a scalar field in an expanding universe.

8.1 Temporal resolution in FRW

In the FRW case, because γ_{eff} decays, the detector does not lose coherence at a constant rate. The bound on temporal resolution is therefore *not* of the form $\Delta\tau \gtrsim (2/\pi\eta)(1+z)^{-1}$ obtained from the static-bath model. It weakens with time, becoming trivial at late times. The informational bound $\Delta\tau \gtrsim T_{\text{total}}\, e^{-I_{\max}/2}$ discussed in the literature of autonomous clocks (26; 27) is a comparative constraint: with $I_{\max} \approx 0.26$ nats it gives a weak numerical bound ($\Delta\tau/T_{\text{total}} \gtrsim 0.88$), not a sharp prediction.

8.2 Reinterpretation of the UAT intuition

The original UAT intuition—that the observer's temporal resolution is bounded by its informational connection to the field—survives the correction. What changes is the specific form of the bound: in FRW, the connection weakens with the expansion, so the bound weakens too. At early times ($z \gtrsim 1$), the detector is strongly coupled to the field; at late times, it is effectively decoupled.

9 What Is Not Claimed

For clarity, this paper does not claim:

1. That time runs at different rates for different observers. The observable is clock resolution, not clock rate.

2. That the metric is modified, or that any cosmological parameter is altered. The background is standard ΛCDM.

3. That the additive formalism of the original UAT is recoverable. It is not. The correct formalism is the modular one.

4. That a power-law decay of clock resolution exists in finite-dimensional unitary systems. It does not.

5. That the relaxation rate $\Gamma = \pi\eta/2$ of Section 6 describes a detector coupled to a cosmological field. It does not. It describes a detector coupled to a static external bath.

6. That the effective damping rate in FRW is constant. It is not; it decays with the expansion and exhibits oscillatory memory corrections.

7. That the first failure of the Markovian approximation in curved spacetime is reported here. It is not. That failure is established in the literature (22; 23). What is new is the 3+1D Gaussian calculation with a full expansion history, the explicit comparison with the Ohmic closure, and the framing via modular information measures.

8. That UAT precedes or anticipates the modular program of **(author?)** (9, 12). It does not. The convergence is independent and parallel; see Section 1.1.

10 Future Work

The extensions of this work are:

1. **Full non-Markovian treatment with analytic regulators.** The numerical computation of $\gamma_{\text{eff}}(\eta)$ uses a sharp UV cutoff. An analytic treatment with the correct distributional prescription would isolate the physical damping from the Lamb shift.

2. **Massive field and conformal coupling.** For $m_\phi \neq 0$ or nonminimal coupling $\xi \neq 1/6$, the mode functions have a different IR behaviour. In particular, the Gibbons–Hawking temperature emerges for $m_\phi \sim H$.

3. **Full ΛCDM background with era transitions.** The present calculation uses power-law $a(\eta)$. The full ΛCDM case with radiation–matter and matter–dark-energy transitions could show whether the era boundaries leave a signature in γ_{eff}.

4. **Relation to the crossed-product algebra and quantum-reference-frame formulations** (8; 10; 12).

11 Conclusions

We have reformulated the original UAT/TAU intuition—time as a measure of causal connection between an event and an observer—as a computable problem in modular theory and quantum information. The reformulation yields:

1. **KMS verification.** The Unruh–DeWitt detector in Minkowski space satisfies the KMS condition at machine precision.

2. **Information mutual.** $I(O{:}\phi)_{\max}$ converges with N to $I_{\max} \approx 0.26$ nats. The power-law exponent α of the Fisher-information decay does not converge with N; the earlier value $\alpha \approx 0.72$ was an artefact.

3. **The Markovian approximation fails in FRW.** A static Ohmic bath gives $\Gamma = \pi\eta/2$, but the exact Wightman function of a scalar field in FRW gives a γ_{eff} that is not constant, does not approach $\pi\eta/2$, and exhibits sign-changing oscillations (memory effects) in the matter and dark-energy regimes. The global decay is robust against the choice of smooth UV regulator.

The comparison between the two continuum models is itself a methodological result: *phenomenological closures of the continuum limit must be derived from the exact Wightman function of the field, not assumed.* The Ohmic bath gives the right answer for the wrong system.

Data and Code Availability

All numerical results in this paper are reproducible from the Python scripts `etapaB_v2.py` (finite-N), `etapaC_v5.py` (static-bath QBM), `frw_wightman.py` (exact FRW Wightman function), and `frw_robustness.py` (robustness against smooth regulators). The scripts use only `numpy`, `scipy`, and `matplotlib`.

Public development record

The UAT/TAU framework was developed independently by the author during 2025. For the reader's reference, the following public records document its development. All timestamps are from the originating platform and are not reconstructed.

- **June 2025.** First public release of the UAT/TAU framework on GitHub: `gh repo clone miguelpercu/tiempo-aplicable-unificado`.
- **11 June 2025.** Release of the equations repository on GitHub: `gh repo clone miguelpercu/ecuaciones`
- **November 2025.** First Zenodo deposits of the UAT/TAU framework:
  - UAT Theory: `10.5281/zenodo.17729221`
  - UPC Principle: `10.5281/zenodo.18210808`
  - Extended Deposit: `10.5281/zenodo.23432466`

These records are provided for completeness. They document an independent development path and are not intended as a priority claim with respect to the modular program.

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